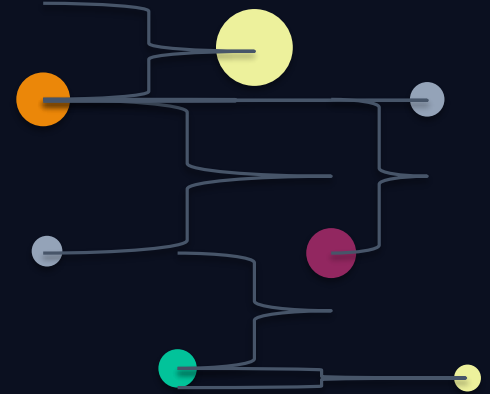


Traffic Accident Patterns in the Canton of Zurich

*An interactive dashboard
for exploring police-recorded traffic accident data*

Thomas Martinetti · Alessandro Pucino · Neil Schranz



THE TOPIC

226'786 accidents — too much for raw tables

- Police-recorded traffic accidents in the Canton of Zurich (2011–2025)
- Source: opendata.swiss
- Each row - one accident event, with type, severity, road type, coordinates, time, and participant flags.

226K

accident records

37

variables

15 yrs

time span

Main goal

Turn a large accident dataset into an interactive dashboard that supports traffic safety planning and insurance risk analysis.

THE QUESTIONS WE SET OUT TO ANSWER

- 1** Where do accidents concentrate across the canton?
- 2** How has accident frequency changed from 2011 to 2025?
- 3** When do accidents peak during the week?
- 4** Which road types and road users show higher accident burden?

Two decision-oriented users, one shared dashboard

The dashboard supports operational traffic safety planning and exploratory insurance risk analysis.

Laura Bianchi

HEAD OF TRAFFIC POLICE OPERATIONS

46 · *high domain expertise · medium data literacy*



NEEDS

Identify accident hotspots, peak time windows and severe road-type patterns

PAIN

Large accident tables are difficult to translate into operational priorities

GAIN

Evidence-based support for monitoring, patrol and prevention discussions

Johannes Schmidt

INSURANCE RISK ANALYST

52 · *medium-high domain expertise · high data literacy*



NEEDS

Explore accident exposure, severity and location patterns for risk review

PAIN

Raw accident records are too detailed for quick portfolio-level interpretation

GAIN

Clear visual comparisons for internal reporting and risk assessment

Dashboard flow — from scale to action

The dashboard guides users from overall accident volume to time patterns, vulnerable road users, severity and spatial concentration. Filters allow users to focus the story on specific years, road types, severities and accident types.

01

Scale:
trend + monthly
drill-down

How has accident
count changed year-
on-year since 2011?

02

Time patterns:
weekday × hour

When do accidents
peak during the
week?

03

Frequency:
accident types

Which accident
types dominate?

04

Severity:
by road type

How does severity
split across different
road types?

05

Space:
accident hotspots

Where do accidents
concentrate?

The final product is interactive: filters, tooltips, drill-downs and hotspot focus options support targeted exploration.

Time patterns become operationally readable

USER PROBLEM

- Different users need different accident subsets.

DASHBOARD SOLUTION

- Sidebar filters define the analytical context.
- KPI cards summarize the selected accident subset.
- All views update consistently from the same filter selection.

VISUAL RHETORIC

- Filters stay on the left, keeping the main page focused on interpretation.
- KPI cards give immediate orientation before users inspect charts.
- The layout follows overview first, then details on demand.

PERSONA VALUE

- Laura can isolate severe accidents, road types or vulnerable users for planning.
- Johannes can compare exposure patterns across selected risk segments.



Traffic Accident Patterns in the Canton of Zurich

Explore police-recorded traffic accidents by time, accident type, road user involvement, severity and location. Use the filters on the left to focus the dashboard on specific years, road types, severities or participant groups.

Selected accidents
226'786

Fatal accidents
416

Severe injury accidents
8'188

Most common road type
Minor road

Most common accident type in the current selection: Accident with skidding or self-accident.

From yearly overview to monthly detail

USER PROBLEM

- Understand **when** accident pressure increases before planning monitoring.
- A **quick way to compare** exposure across **years and months**.

DASHBOARD SOLUTION

- The yearly line chart gives the **long-term** overview.
- The **monthly drill-down** keeps detail available without overloading the main chart.
- Hover **tooltips** provide **exact values** only when needed.

VISUAL RHETORIC

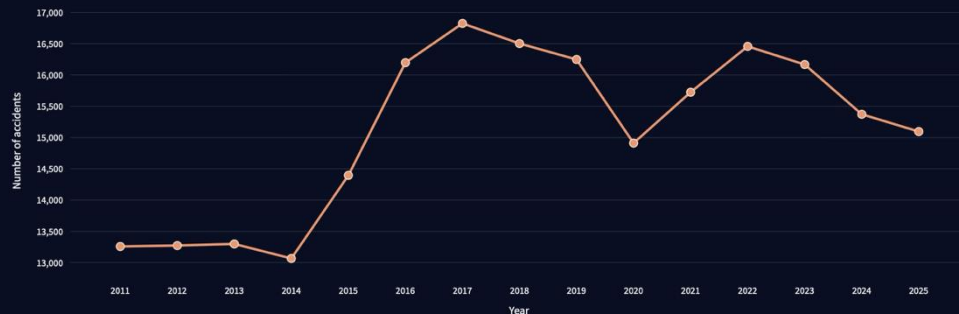
- Line + points make change over time **easy to follow**.
- Drill-down supports **“overview first, details on demand”**.
- A **single accent colour** keeps attention on temporal development.

PERSONA VALUE

- Laura moves **from annual** pressure **to seasonal** planning periods.
- Johannes gets **time-based evidence** for risk review.

Accident trend over time

Yearly totals show the long-term development. The monthly view highlights seasonal variation within a selected year.



Who is involved with what happens?

USER PROBLEM

- Laura needs to plan **safety actions for vulnerable groups**.
- Johannes needs to **compare accident exposure** by participant group.

DASHBOARD SOLUTION

- One chart **compares** pedestrian, bicycle and motorcycle **involvement**.
- The second chart **splits frequent accident types** by vulnerable road user group.
- **Consistent colours** connect the **same groups** across both views.

VISUAL RHETORIC

- Simple bars give a **fast group comparison**.
- Stacked horizontal bars **connect accident type** and **road user** involvement.
- Horizontal layout keeps **long accident type labels** readable.

PERSONA VALUE

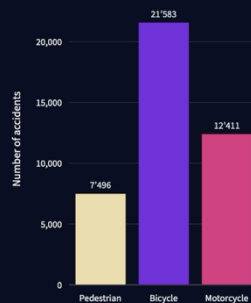
- Laura can **support** targeted prevention **campaigns**.
- Johannes can **compare risk profiles** by **road user** group.

Deploy

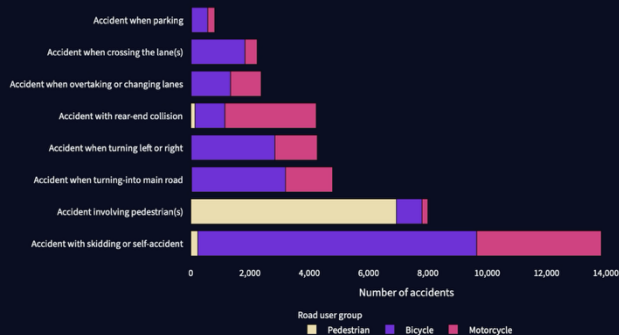
Vulnerable road users and accident types

The first view shows vulnerable road user involvement. The second breaks frequent accident types down by road user group.

Vulnerable road users



Accident types by vulnerable road user



04 - HOW SEVERE ARE ACCIDENTS BY ROAD TYPE?

Severity becomes comparable across road environments

USER PROBLEM

- Laura needs to **distinguish high-volume road types** from **more serious** outcomes.
- Johannes needs **severity information** for risk interpretation.

DASHBOARD SOLUTION

- Stacked bars show total **accident burden** and **severity** mix at the same time.
- Legend **interaction** lets users focus on **one severity category**
- **Tooltips** provide count and share without cluttering the chart.

VISUAL RHETORIC

- Bar height communicates **overall burden**.
- **Colour** separates **severity levels consistently**.
- Stacking supports comparison of both **volume** and **composition**.

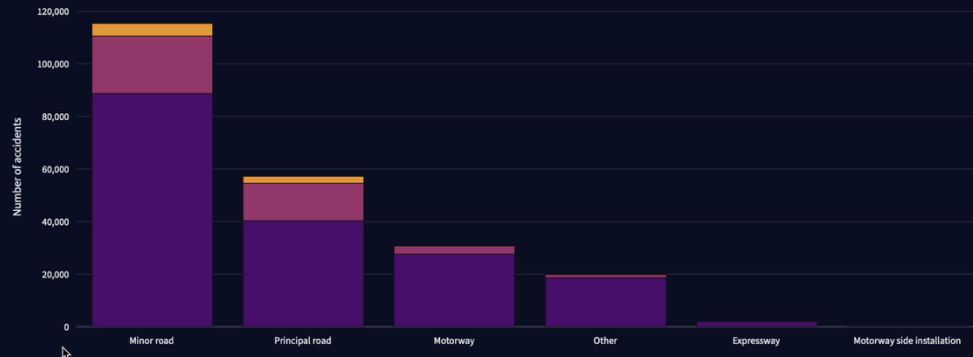
PERSONA VALUE

- Laura can **prioritize road-safety** discussions by road type.
- Johannes can **compare severity** exposure across road environments.

04 - HOW SEVERE ARE ACCIDENTS BY ROAD TYPE?

Accident severity by road type

The stacked bars compare total accident volume and severity mix across road types.



Coordinates become spatial decision evidence

USER PROBLEM

- Laura needs to know **where** monitoring or prevention discussions should **focus**.
- Johannes needs to **identify** where accident **exposure** is spatially concentrated.

DASHBOARD SOLUTION

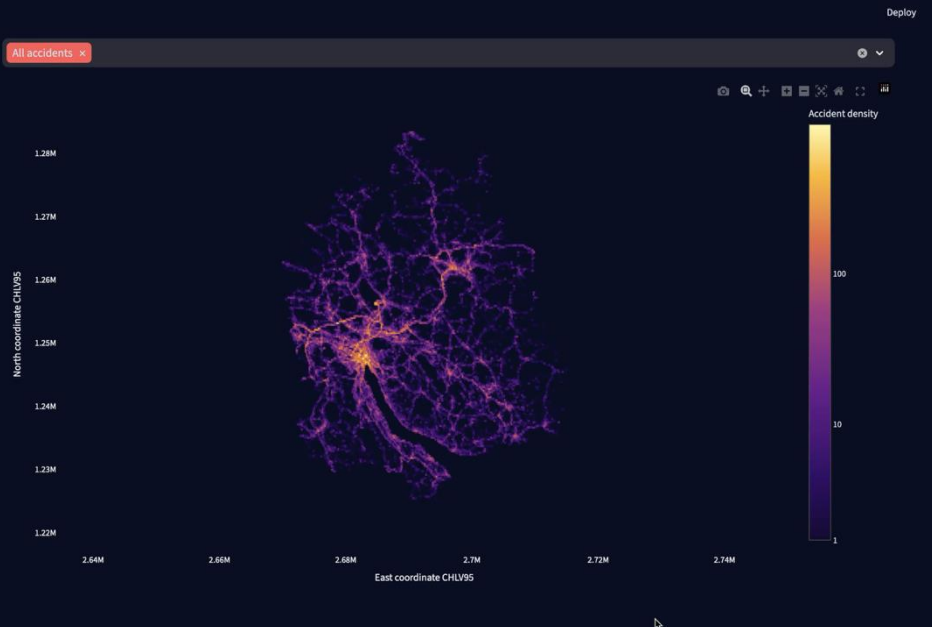
- **Accident coordinates** are aggregated into **density cells**.
- **Zooming** allows users to inspect dense areas more closely.
- **Hotspot focus** options **isolate** severe, fatal or vulnerable road user accidents.

VISUAL RHETORIC

- **Density aggregation** makes thousands of points readable.
- **Colour intensity** communicates **concentration**.
- **Spatial layout** preserves the **geography** of the accident pattern.

PERSONA VALUE

- Laura can identify **candidate areas** for traffic **safety** planning.
- Johannes can **locate spatial exposure** clusters for risk review.



One Dashboard. Fully reproducible.

The product is an interactive Streamlit dashboard, supported by Quarto documentation. Every chart, number, and data-cleaning step can be traced back to the original open data source.

This matters for both users. Laura can use the dashboard as evidence for traffic safety planning, while Johannes can trust that the risk patterns are based on official police data.

*Dashboard published via Streamlit Community Cloud —
documentation published via GitHub Pages.*

Source: opendata.swiss · Polizeilich registrierte Verkehrsunfälle im Kanton Zürich seit 2011

Streamlit

Interactive dashboard with filters, tooltips, monthly drill-down and hotspot exploration.

Quarto + GitHub

Documentation, version control, data report, design rationale and deployment notes.

Open data

Raw CSV from the official cantonal registry — no scraping, no proprietary sources.

Where we need your input

1 Spatial clarity

Can users identify the main accident hotspot within a few seconds?

2 Temporal clarity

Can users find peak weekday-hour windows and inspect monthly patterns?

3 Filter usability

Can users focus the dashboard by year, severity, road type, accident type or road user group without confusion?

4 Decision relevance

Does the dashboard support Laura's planning questions and Johannes' risk review questions?

5 Scope & prioritisation

Are the views enough to support the use cases without overloading the interface?

Thank you for the attention



We are happy to answer your questions

